

Chrono-Acoustic Brane Mechanics (CABM) v 0.2”

“Unifying Dark Matter and Dark
Energy via Superluminal Ontological
Tension and Novikov Self-Consistency”

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Abstract

We propose a non-particulate resolution to the
dark matter and dark energy puzzles by

redefining the energy-momentum tensor through an **Ontological Tension Tensor** $\Phi_{\mu\nu}$. Standard 4D spacetime is modeled as a boundary brane embedded within an N-dimensional bulk where superluminal field phases propagate. Drawing upon Einstein's geometric formulations, Penrose's singularity-free topological transitions, and Dragan's superluminal spacetime transformations, we demonstrate that galactic rotation curves and cosmic acceleration are causal-retrocausal reflections of future boundary states. By enforcing the Novikov self-consistency principle upon continuous multi-dimensional fields, we eliminate causal paradoxes. Furthermore, we incorporate a **Chrono-Acoustic framework** where baryonic matter emerges as localized standing-wave harmonics generated by a fundamental bulk frequency Ω_{bulk} vibrating the spacetime membrane. The macro-metric collapses into

stable 4D locality via observer-induced decoherence. We formulate a definitive observational test using Pulsar Timing Arrays (PTAs) to isolate the predicted phase-flicker noise $h_c(f) \propto f^{-1/3}$.

I. Introduction

The Λ CDM paradigm, while observationally robust at cosmological scales, requires the introduction of two unobserved dark components: cold dark matter (CDM) and a finely-tuned cosmological constant (Λ). Despite decades of underground detection efforts, directly interacting dark matter candidates (e.g., WIMPs) remain elusive.

Concurrently, attempts to modify Einstein's

General Relativity at low accelerations often lack a fundamental holographic or multi-dimensional justification. In this paper, we synthesize the superluminal coordinate extensions proposed by Dragan et al. with Penrose's twistor and cosmic censorship frameworks. We show that what is perceived as dark matter and dark energy are not physical substances, but localized and global manifestations of an **Ontological Tension** $\Phi_{\mu\nu}$ generated by the back-propagation of future hyper-dimensional wavefunctions. Causal integrity is strictly preserved via a field-theoretic extension of Novikov's self-consistency principle.

II. Mathematical Framework

Let our physical universe be a 4-dimensional

Lorentzian brane M_4 with metric $g_{\mu\nu}$ embedded in an N -dimensional bulk M_N . The modified Einstein field equations are:

$$G_{\mu\nu} + \Lambda_0 g_{\mu\nu} = 8\pi G c^4 (T_{\mu\nu}^{\text{baryon}} + \Phi_{\mu\nu})$$

$$= c^4 8\pi G (T_{\mu\nu}^{\text{baryon}} + \Phi_{\mu\nu})$$

where Λ_0 is the bare vacuum energy and $\Phi_{\mu\nu}$ is the **Ontological Tension Tensor**. Following the Gauss–Codazzi equations, $\Phi_{\mu\nu}$ arises from the extrinsic curvature tensors $K_{\mu\nu}$ and the projection of the bulk Weyl tensor $E_{\mu\nu}$:

$$\Phi_{\mu\nu} = -\frac{1}{2} E_{\mu\nu} + \alpha \Pi_{\mu\nu}$$

The tension tensor is governed by an advanced Green's function integrating over future hyper-volumes:

$$\Phi_{\mu\nu}(x) = \int_{-\infty}^t d\tau \int dN -4 X G_{\mu\nu}(x, X, \tau) \nabla_A \Psi^{\text{bulk}}(\tau, X^A)$$

$$\Phi_{\mu\nu}(x) = \int_{-\infty}^t d\tau \int dN -4 X G_{\mu\nu}(x, X, \tau) \nabla_A \Psi^{\text{bulk}}(\tau, X^A)$$

ER bridge current with bulk exchange:

$$\nabla_\mu \Phi_{\mu\nu} = J_\nu^{\text{ER}} - \nabla_\mu T_{\mu\nu}^{\text{bulk}}$$

$$\nabla_\mu \Phi_{\mu\nu} = J_\nu^{\text{ER}} - \nabla_\mu T_{\mu\nu}^{\text{bulk}}$$

Chrono-acoustic wave equation:

$$(\Box^4 - \Omega_{\text{bulk}}^2 v s^2) \Phi_{\mu\nu} = \kappa |\Psi_{\text{bulk}}|^2 g_{\mu\nu} + \alpha T_{\mu\nu}^{\text{baryon}}$$

$$(\Box^4 - v s^2 \Omega_{\text{bulk}}^2) \Phi_{\mu\nu} = \kappa |\Psi_{\text{bulk}}|^2 g_{\mu\nu} + \alpha T_{\mu\nu}^{\text{baryon}}$$

Novikov self-consistency fixed point:

$$\Phi_{\mu\nu}(t) = P^{\text{adv}}[U^{\text{}}(\Phi_{\mu\nu}(t) + T_{\mu\nu}^{\text{baryon}}(t))] \equiv F[\Phi]$$

$$\Phi_{\mu\nu}(t) = P^{\text{adv}}[U^{\text{}}(\Phi_{\mu\nu}(t) + T_{\mu\nu}^{\text{baryon}}(t))] \equiv F[\Phi](t)$$

Observer-induced metric stabilization:

$$g_{\mu\nu}(x) = \langle \Psi_{\text{obs}} | G^{\mu\nu} | \Psi_{\text{obs}} \rangle \quad g_{\mu\nu}(x) = \langle \Psi_{\text{obs}} | G^{\mu\nu} | \Psi_{\text{obs}} \rangle$$

Singularity-free black hole core:

$$\lim_{r \rightarrow 0} \Phi_{\mu\nu} = -8\pi G c^4 T_{\mu\nu}^{\text{baryon}} \quad \lim_{r \rightarrow 0} \Phi_{\mu\nu} = -c^4 8\pi G T_{\mu\nu}^{\text{baryon}}$$

Predicted PTA signature:

$$h_c(f) \propto f^{-1/3} \quad h_c(f) \propto f^{-1/3}$$

III. Axioms and Theorems

Axioms

- **A1.** $(F, +, \cdot, >)$ is an ordered field with $1 > 0$.
- **A2.** $M_4 \hookrightarrow M_N, N \geq 5$, signature $(-, +, \dots, +)$.
- **A3.** Induced metric: $g_{\mu\nu} = \partial_{\mu} X^A \partial_{\nu} X^B G_{AB}$.
- **A4.** $\Phi_{\mu\nu} u^{\mu} u^{\nu} > 0$ for every timelike u^{μ} .
- **A5.** Novikov fixed point $\Phi = F[\Phi]$ has exactly one solution.
- **A6.** Chrono-Acoustic closure: $T^{\text{baryon}}_{\mu\nu} = T_{\mu\nu}[\Phi]$.

Theorems

T1. In any ordered field F , $1 > 0$.

Proof. Suppose $1 < 0$. Then $-1 > 0$. But $(-1)(-1) = 1$, and the product of two positive elements is positive, so $1 > 0$. Contradiction. Hence $1 > 0$. ■

T2. F has characteristic 0 and contains a copy of $\mathbb{Q}$.

T3. If the Novikov operator F is a contraction with

constant $L < 1$, the self-consistent field Φ is unique (Banach fixed-point theorem).

T4. In $d = 3$, topological phase fluctuations of Φ yield $S_{\Phi}(f) \propto f^{-(d-2)/d} = f^{-1/3}$.

IV. Novikov Self-Consistency and Paradox Resolution

The grandfather paradox is not forbidden by fiat; it is topologically unrepresentable. The bulk manifold M_N admits only self-consistent field configurations. The phantom weight felt by a galaxy today is the exact gravitational anchor required to form the future galactic structures that emit that very echo. Causal loops are fixed points, not violations.

V. Chrono-Acoustic Framework

V.1 Quantum-Topological Flux via ER Bridges

Dark matter is a gravitational shadow of energy flowing from higher-dimensional manifolds into the 4D brane through micro-wormholes.

Following $ER = EPR$, this flux is a current density J^{ER}_ν driving the non-vanishing divergence of $\Phi_{\mu\nu}$.

V.2 The Cosmic Sound

Baryonic matter is not rigid point-particles but localized standing-wave harmonics of Ω_{bulk} vibrating the spacetime membrane. Mass density emerges via a modified non-linear Helmholtz

equation.

V.3 Observer-Induced Stabilization

The global bulk field is a superposition of infinite probabilistic geometries. The illusion of rigid 4D locality is a relational artifact of decoherence. Consciousness is not required; any interaction suffices.

VI. Singularity-Free Black Holes

As $r \rightarrow 0$, extreme matter density triggers non-linear growth in Φ . The future-state boundary condition enforces a negative pressure profile, so g_{00} does not diverge. The metric transitions smoothly into a regular chrono-topological core, fulfilling Penrose's requirement.

VII. Observational Predictions — PTA

Because $\Phi_{\mu\nu}$ is an integrated projection of superluminal phase fields, it cannot be perfectly smooth. It must exhibit a quantum-topological flicker.

We predict a characteristic strain signature:

$$h_c(f) \propto f^{-1/3} \quad h_c(f) \propto f^{-1/3}$$

This cannot be replicated by CDM particles or MOND. The decisive test is IPTA DR3 (2026–2027): measure the tilt β of $h_c(f)$. SMBHB predicts $\beta = -2/3$. CABM predicts $\beta = -1/3$.

Comparison with NANOGrav 15 yr

f (nHz)	NANOGrav h_c	CABM (Case C)	CABM (Case A)
2	6.27	2.52	0.40
31.7	1.00	1.00	1.00
100	0.465	0.683	1.47

Case C ($h_c \propto f^{-1/3}$) lies within 40–150% of NANOGrav across the PTA band. Case A is excluded.

VIII. Numerical Model

1+1D scalar field $\phi(t,x)$ on the brane with self-interaction and advanced (Novikov) source:

$$\partial_t^2\phi-\partial_x^2\phi+m^2\phi+\lambda\phi^3=J(x,t)\partial_t^2\phi-\partial_x^2$$

$$\varphi + m^2 \varphi + \lambda \varphi^3 = J(x, t)$$

Advanced source:

$$J(x, t) = \beta \int_{-\infty}^t dt' G_{\text{adv}}(t, t') \varphi^3(x, t') \quad J(x, t) = \beta \int_{-\infty}^t dt' G_{\text{adv}}(t, t') \varphi^3(x, t')$$

Discretization:

$$\varphi_{ij+1} = 2\varphi_{ij} - \varphi_{ij-1} + \Delta t^2 [\partial_x^2 \varphi_{ij} - m^2 \varphi_{ij} - \lambda (\varphi_{ij})^3 + J_{ij}]$$

$$\varphi_{ij+1} = 2\varphi_{ij} - \varphi_{ij-1} + \Delta t^2 [\partial_x^2 \varphi_{ij} - m^2 \varphi_{ij} - \lambda (\varphi_{ij})^3 + J_{ij}]$$

Contraction criterion:

$$L \approx 3\beta \tau_{\text{bulk}} A^2 V, L < 1 \Leftrightarrow \beta < 1/3 \tau_{\text{bulk}} A^2 V \quad L \approx 3\beta \tau_{\text{bulk}} A^2 V, L < 1 \Leftrightarrow \beta < 3 \tau_{\text{bulk}} A^2 V$$

Expected results:

- Stable soliton for $\lambda > 0$, $m^2 > 0$.
- Unique fixed point for $L < 1$.
- Bifurcation at $L \rightarrow 1^-$: two branches — signal of multi-history topology.
- Spectral exponent: $1D \rightarrow \alpha \approx 1$; $3D \rightarrow \alpha \approx 1/3$ (topological defect scaling).

IX. Synthesis — $1 \succ 0$

$1 \succ 0 \implies \Phi_{\mu\nu} u^\mu u^\nu \succ 0 \implies \text{Ontological Tension}$
 $\implies \text{Geometry} \implies \text{Matter} \implies S_\Phi(f) \propto f^{-1/3} \implies$
PTA test.

$1 \succ 0$ generates tension; tension generates geometry; geometry generates matter; matter generates a measurable flicker in spacetime.

X. Conclusion

CABM v0.2 provides (i) axioms, (ii) a falsifiable PTA prediction, (iii) a numerical solver, and (iv) a natural origin for the $-1/3$ exponent. It is a working framework, not a claim of discovery. The decisive test is IPTA DR3.

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